

P2848R0

std::is_uniqued

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3

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rawgit.com/Quuxplusone/draft/gh-pages/d2848-std-is-uniqued.html

Abstract

The STL provides the triples (`std::sort`, `std::is_sorted_until`, `std::is_sorted`) and (`std::make_heap`, `std::is_heap_until`, `std::is_heap`) and (`std::unique`, `std::adjacent_find`, [blank]). We fill in the blank by proposing `std::is_uniqued`.

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§ 1. Changelog

- R0:
 - Initial revision.

§ 2. Motivation and solution

The "classic STL" provides the following sets of related algorithms:

- `is_sorted_until` (returns iterator), `is_sorted` (i.e. `is_sorted_until == end()`)
- `is_heap_until` (returns iterator), `is_heap` (i.e. `is_heap_until == end()`)
- `adjacent_find` (returns iterator), **[blank]** (i.e. `adjacent_find == end()`)
- `mismatch` (returns iterators), `equal` (i.e. `mismatch == {end(), end()}`)
- `find_if` (returns iterator), `none_of` (i.e. `find_if == end()`)
- `find` (returns iterator), `contains` (i.e. `find != end()`)
- `search` (returns iterator), `contains_subrange` (i.e. `search != end()`)

And these, too:

- `sort` (i.e. make the input range satisfy `is_sorted`)

- `make_heap` (i.e. make the input range satisfy `is_heap`)
- `unique` (i.e. make the input range satisfy **[blank]**)

The missing algorithm is what would fill in the blank in the following invariant assertions:

```
std::set<K> s = { ... };
std::multiset<K> ms = { ... };
std::map<K, V> m = { ... };
std::multimap<K, V> mm = { ... };
std::unordered_set<K> us = { ... };
std::unordered_map<K, V> um = { ... };

template<class M, class P = M::value_type>
auto ValueEq(const M& m) {
    return [eq = m.key_eq()](const P& a, const P& b) {
        return eq(a.first, b.first);
    };
}

assert(std::is_sorted(s.begin(), s.end(), s.value_comp()));
assert(std::is_sorted(ms.begin(), ms.end(), ms.value_comp()));
assert(std::is_sorted(m.begin(), m.end(), m.value_comp()));
assert(std::is_sorted(mm.begin(), mm.end(), mm.value_comp()));
assert(std::is____ed(s.begin(), s.end(), std::not_fn(s.value_comp())));
assert(std::is____ed(m.begin(), m.end(), std::not_fn(m.value_comp())));
assert(std::is____ed(us.begin(), us.end(), us.key_eq()));
assert(std::is____ed(um.begin(), um.end(), ValueEq(um)));
```

We propose that this algorithm should exist, and should be spelled `std::is_uniqued`.

§ 2.1. Prior art in other languages

The C++ STL verb `std::unique` is derived from Unix's `uniq` filter, which filters adjacent matching lines from the input.

The Swift language provides a transformation named `sorted` (corresponding to C++'s `std::sort`), and a transformation named `uniqued`. The latter doesn't quite correspond to the C++/Unix notion of "uniquing": it removes even non-adjacent duplicates, by putting all the elements into a hashset and

then pulling them out again. Still, this is prior art for the idea of treating "unique, uniqued, uniquing" as a verb.

```
let animals = ["dog", "pig", "cat", "ox", "dog", "cat"]
let u = Array(animals.uniqued())
// 'u' is now ["dog", "pig", "cat", "ox"]
```

§ 2.2. Should we require `pred` to be an equivalence relation?

Currently, `std::unique` has undefined behavior if you pass it a predicate that is not an *equivalence relation*. Fortunately, `std::adjacent_find` does not.

For example, the author of `std::flat_set` might write:

```
container_type c;
key_compare compare;
std::sort(c.begin(), c.end(), compare);
c.erase(std::unique(c.begin(), c.end(), std::not_fn(compare)), c.end()); //
assert(std::is_sorted(c.begin(), c.end(), compare));
assert(std::adjacent_find(c.begin(), c.end(), std::not_fn(compare)) == c.er
```

Both before and after this proposal, the line marked "UB" has undefined behavior (although it will generally work in practice). We don't propose to change the precondition of `std::unique`. The line marked "OK" has well-defined behavior. we don't propose to change the precondition of `std::adjacent_find` either.

After this proposal, the following shorter line will be equivalent to the line marked "OK":

```
assert(std::is_uniqued(c.begin(), c.end(), std::not_fn(compare))); // OK
```

That is, the new algorithm `std::is_uniqued` will have the same precondition as the existing `std::adjacent_find` algorithm (in terms of which it is defined). This differs from the precondition of `std::unique`, which is a little surprising, but we believe it's the correct choice.

§ 3. Implementation experience

Arthur has implemented `is_unique` and `ranges::is_unique` in his fork of libc++; see [commit 490536e](#). This implementation is [available on Godbolt Compiler Explorer](#).

§ 4. Proposed wording relative to C++23

Add a new section after [\[alg.adjacent.find\]](#).

NOTE: This wording is copied straight from [\[is.sorted\]](#), with these mechanical replacements: `is_sorted` becomes `is_unique`, `is_sorted_until` becomes `adjacent_find`, `Compare comp` becomes `BinaryPredicate pred` (cf. [\[alg.adjacent.find\]](#)), `indirect_strict_weak_order` becomes `indirect_binary_predicate` (ditto), and `ranges::less` becomes `ranges::equal_to` (ditto).

§ 4.1. `is_unique` [\[alg.is.unique\]](#)

```
template<class ForwardIterator>
constexpr bool is_uniqued(ForwardIterator first, ForwardIterator last);
```

Effects: Equivalent to: return adjacent_find(first, last) == last;

```
template<class ExecutionPolicy, class ForwardIterator>
bool is_uniqued(ExecutionPolicy&& exec,
               ForwardIterator first, ForwardIterator last);
```

Effects: Equivalent to: return adjacent_find(std::forward<ExecutionPolicy>(exec), first, last) == last;

```
template<class ForwardIterator, class BinaryPredicate>
constexpr bool is_uniqued(ForwardIterator first, ForwardIterator last,
                        BinaryPredicate pred);
```

Effects: Equivalent to: return adjacent_find(first, last, pred) == last;

```
template<class ExecutionPolicy, class ForwardIterator, class BinaryPredicate>
bool is_uniqued(ExecutionPolicy&& exec,
               ForwardIterator first, ForwardIterator last,
               BinaryPredicate pred);
```

Effects: Equivalent to: return adjacent_find(std::forward<ExecutionPolicy>(exec), first, last, pred) == last;

```
template<forward_iterator I, sentinel_for<I> S, class Proj = identity,
        indirect_binary_predicate<projected<I, Proj>,
                                   projected<I, Proj>> Pred = ranges::equal_to>
constexpr bool ranges::is_uniqued(I first, S last, Pred pred = {}, Proj proj = {});
template<forward_range R, class Proj = identity,
        indirect_binary_predicate<projected<iterator_t<R>, Proj>,
                                   projected<iterator_t<R>, Proj>> Pred = ranges::equal_
constexpr bool ranges::is_uniqued(R&& r, Pred pred = {}, Proj proj = {});
```

Effects: Equivalent to: return ranges::adjacent_find(first, last, pred, proj) == last;